Lesson Objective

Plot data on line graphs and analyze trends.

Materials

(S) Activity 1 Worksheet

PROBLEM

1 **Read and interpret a line graph.**

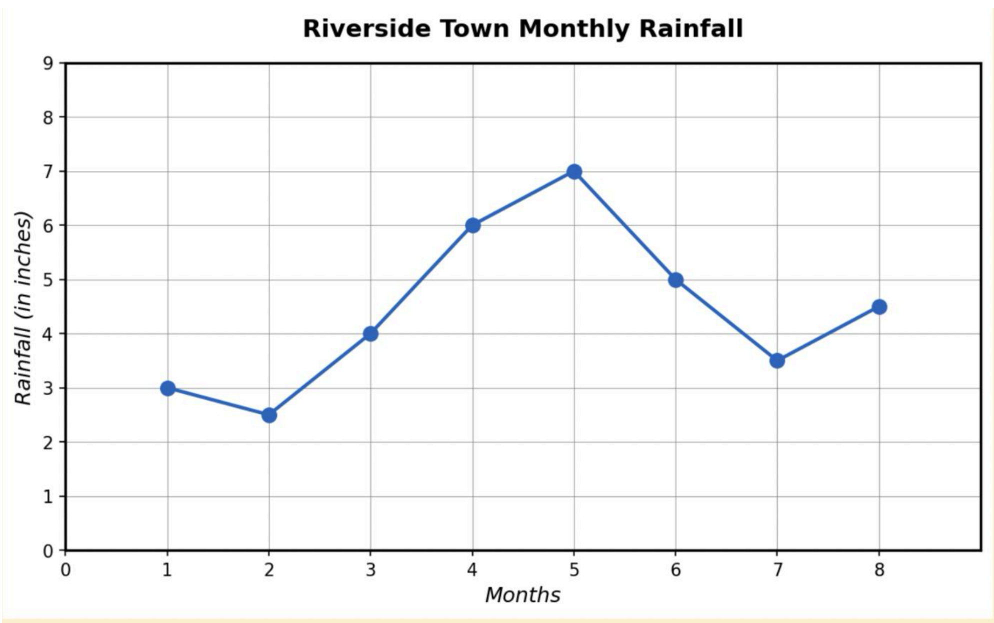
The line graph below tracks the total rainfall in Riverside Town, measured each month, for 8 months. Use the information in the graph to answer the questions that follow.

a. How many inches of rain fell in Month 3?
4 inches.

b. Which month had the greatest amount of rainfall?
Month 5.

c. There was a large increase in rainfall between Month 4 and Month 5. During what other periods did rainfall increase?
Months 2-4 and 7-8.

d. How much more rainfall did the town receive in Month 5 compared to Month 2?
3.5 inches.



TEACHER GUIDANCE

Guide students through reading and analyzing the line graph.

Together, identify the quantity shown on each axis and the scale on each axis. Discuss what the segments on the graph represent in the context of the problem, such as intervals that are constant, increasing, or decreasing.

Then, answer the questions together, asking students to justify each answer.

PROBLEM	TEACHER GUIDANCE

2 **Read and interpret a line graph.**

Students work independently.

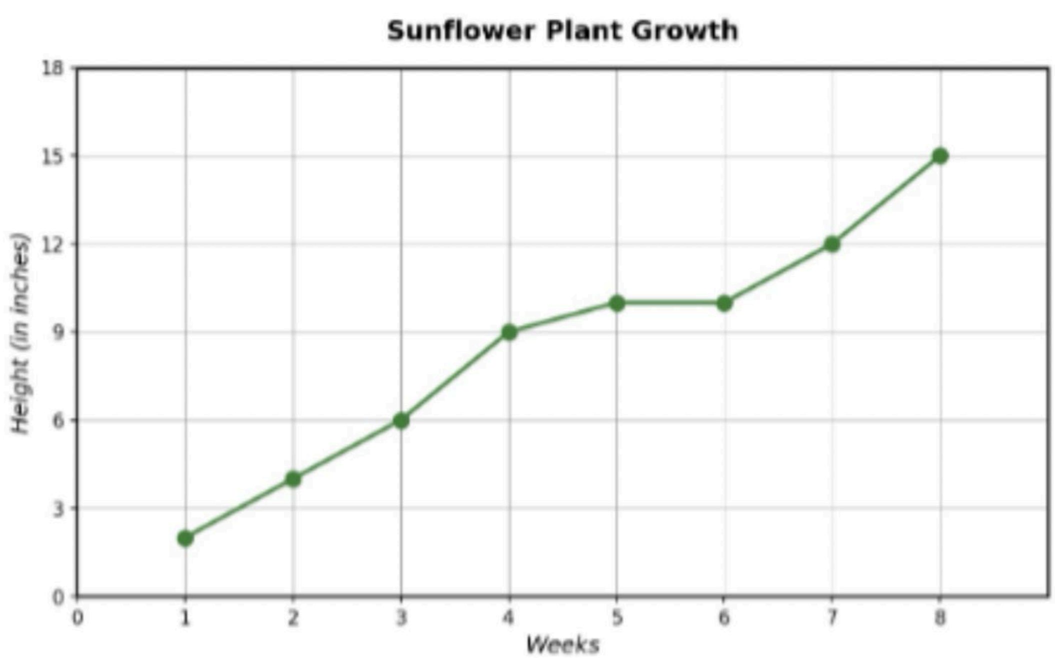
The line graph below tracks the height of a sunflower plant, measured each Sunday, for 8 weeks. Use the information in the graph to answer the questions that follow.

a. About how many inches tall was the plant in Week 2?
About 4 inches.

b. Between which two weeks did the plant grow the most?
Weeks 3-4 or 7-8.

c. The plant did not grow at all between weeks 5 and 6. When else did the plant have very little growth? Explain why that might have happened.
Weeks 4-5. Maybe there was not enough rain or sunlight.

d. What might have caused the rapid growth between weeks 7 and 8?
Maybe it was very sunny and/or rainy.



Monitor For

- Do students have difficulty determining the segments of the graph that are increasing, decreasing, or constant?
- Do students answer with the wrong coordinate (e.g., reading from the x-axis instead of the y-axis or vice versa)?
- Do students need additional support finding the coordinates for points that are not on a clear intersection of the x- and y-axes?

Instructional Moves

- Show students that when we read a graph from left to right, a segment that goes up is increasing, a segment that goes down is decreasing, and a segment that is horizontal shows no change (constant).
- Remind students to pay attention to the units shown on each axis.
- Draw additional lines or labels to each axis to help students determine the coordinates.

Reflection Questions

1. How did you know when it rained the most or least?
I looked for the highest and lowest values on the line graph.

2. How did you know when the amount of rain was increasing or decreasing?
When the line segment slants up from left to right, that's increasing. When it slants down, that's decreasing.

3. How did you analyze and interpret the graph? What parts of the graph did you use?
Answers vary. What parts of the graph did you use? Answers vary and may include: axes labels, quantities, title, points on the graph.